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The Impact of Ice Formation on Wind Turbine Performance and Aerodynamics

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Abstract

Wind energy is the world's fastest growing source of electricity production; if this trend is to continue, sites that are plentiful in terms of wind velocity must be efficiently utilized. Many such sites are located in cold, wet regions such as the Swiss Alps, the Scandinavian coastline, and many areas of China and North America, where the predicted power curves can be of low accuracy, and the performance often deviates significantly from the expected performance. There are often prolonged shutdown and inefficient heating cycles, both of which may be unnecessary. Thus, further understanding of the effects of ice formation on wind turbine blades is required. Experimental and computational studies are undertaken to examine the effects of ice formation on wind turbine performance. The experiments are conducted on a dynamically scaled model in the wind turbine test facility at ETH Zurich. The central element of the facility is a water towing tank that enables full-scale nondimensional parameters to be more closely matched on a subscale model than in a wind tunnel. A novel technique is developed to yield accurate measurements of wind turbine performance, incorporating the use of a torquemeter with a series of systematic measurements. These measurements are complemented by predictions obtained using a commercial Reynolds-Averaged Navier-Stokes computational fluid dynamics code. The measured and predicted results show that icing typical of that found at the Guetsch Alpine Test Site (2330 m altitude)

can reduce the power coefficient by up to 22% and the annual energy production (AEP) by up to 2%. Icing in the blade tip region, 95–100% blade span, has the most pronounced effect on the wind turbine's performance. For wind turbines in more extreme icing conditions typical of those in Bern Jura, for example, icing can result in up to 17% losses in AEP. Icing at high altitude sites does not cause significant AEP losses, whereas icing at lower altitude sites can have a significant impact on AEP. Thus, the classification of icing is a key to the further development of prediction tools. It would be advantageous to tailor blade heating for prevention of ice buildup on the blade's tip region. An "extreme" icing predictive tool for the project development of wind farms in regions that are highly susceptible to icing would be beneficial to wind energy developers.

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